Separating coverage gains from error targeting in molecular prediction sets

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Abstract

Molecular screening benefits from prediction sets that recover errors while preserving useful decisions at a manageable deferral workload. We introduce a class-budget comparison that isolates error targeting by fixing the predictor, eligible targets, and expansion count within each original singleton class. We also derive an exact weighted-rank activation condition and show when constant query weights make expansion equivalent to omitted-label probability ranking. The evaluation contains 960 predictor units across twelve Tox21 assays and random/scaffold holdouts from four additional assay families. On the historical six-assay EPA panel with a stacked predictor, omitted-label ranking achieved 85.10% minimum-class coverage and 46.57% macro correct-singleton yield, compared with 84.75% and 44.61% for generic probability uncertainty at the same 44.02% deferral rate. It also reproduced TAME's 1.3649-percentage-point coverage gain at the same class-specific budget; the TAME-minus-ranking coverage difference was 0.000 points (molecule-cluster bootstrap 95% interval, -0.028 to 0.029). The activation analysis identified a distinct mechanism: both transport gates passed in all 30 primary active-label cells, yet no additional label was added. Class-budget matching thus supplies a decision-aligned reference, while the activation condition connects transport weights to actual interventions. Together they provide a framework for measuring the information gained from molecular prediction-set expansion.

Keywords: molecular prediction; conformal prediction; distribution shift; selective prediction; deferral; class-specific coverage

1 Introduction

Molecular property prediction supports decisions about which compounds to advance, discard, or measure again. A binary prediction set makes the distinction explicit: a singleton selects one outcome, while a two-label set leaves the decision unresolved. Expanding singletons into two-label sets can protect measured outcomes that would otherwise be excluded. It also increases the number of compounds requiring further

evidence. Evaluating this trade-off requires asking both how often a system covers the measured outcome and which decisions it withholds.

Conformal prediction constructs sets with coverage guarantees under specified sampling assumptions. Marginal coverage and class-conditional coverage answer different questions; Mondrian calibration addresses the latter under within-class exchangeability [1]. Molecular collections often differ in composition and support, motivating methods that use unlabeled target compounds to estimate distribution shift. Weighted conformal prediction formalizes likelihood-ratio adjustment under covariate shift [2]. CoDrug combines representation learning with density estimation [3], while KMM-CP uses kernel mean matching and selective support overlap [4]. Their ability to restore coverage is a different question from their ability to make informative decisions at a given workload.

Selective classification has long studied the relation between error and abstention [11]. For a prediction-set expansion, however, matching only total abstention can leave a further imbalance: methods may defer different proportions of original class-0 and class-1 decisions. This matters in molecular assays with uneven class frequencies and unequal costs of false-positive and false-negative decisions. Moreover, a Mondrian singleton need not be the probability argmax. Ranking uncertainty around probability 0.5 may therefore differ from ranking the probability of the particular label omitted by that singleton.

We use the transport-audited multi-view envelope (TAME) as a concrete case study. TAME adds a protected label when two weighted views agree, subject to transport diagnostics. Its historical label-0 version improved minimum-class coverage relative to a stacked Mondrian baseline. The interpretation depends on whether the selected deferrals contain more errors than an equally costly simple rule. We examine this directly by matching the expansion count within each original singleton class and ranking the probability of the omitted label.

The study makes three connected contributions. First, class-budget matching and an error–yield decomposition isolate error targeting from expansion volume and predicted-class allocation. Second, an exact weighted-rank condition links transport weights to label addition and identifies when expansion reduces to probability ranking. Third, a 960-unit molecular evaluation demonstrates the value of a decision-aligned reference: omitted-label ranking achieves higher coverage and correct-singleton yield than generic uncertainty at the historical TAME budget and reproduces the transport envelope's coverage gain. The prespecified active-label comparison, alternative predictors and scores, and four additional assay families establish how eligibility, intervention, and error recovery determine these outcomes.

2 Methods

2.1 Molecular cohorts and experimental roles

The Tox21 challenge supplies twelve nuclear-receptor and stress-response assays [6]. The source collection supplies model-fitting, stack-fitting, and calibration roles. The public leaderboard contains 296 structures and the final EPA collection 647 before structure standardization and overlap exclusions. The historical primary panel comprises NR-AhR, NR-ER, SR-ARE, SR-ATAD5, SR-MMP, and SR-p53, selected in the original development analysis by at least twenty positive labels. We retain this panel and report all twelve

assays separately. EPA labels had been accessed in the earlier study, so the present EPA comparisons are retrospective.

The additional assay families are AMES mutagenicity, the standard hERG blockade dataset, drug-induced liver injury (DILI), and CYP2D6 inhibition from the Veith collection [7–10]. Tox21 labels denote receptor or stress-response activity; the additional datasets retain their endpoint-specific label definitions. Standardization uses RDKit 2026.03.4. Same-label duplicate structures are collapsed, and structures with conflicting binary labels are removed. The resulting AMES, hERG, DILI, and CYP2D6 source tables contain 7,255, 607, 474, and 13,083 structures, respectively.

Each additional dataset has a fixed random target split and a fixed scaffold target split, each approximately 20% of the source table. Target counts are 1,451, 122, 95, and 2,617, respectively. Outer seeds are 43021 and 43022. Target selection uses structure or scaffold identity without outcome stratification. Five fits, seeds 311–315, reuse each fixed target cohort. These are within-dataset holdouts across additional endpoints. Supplementary Table S1 reports target, labeled-class, and calibration counts; acquisition and exclusion rules are specified in the archived data adapters.

2.2 Predictors and prediction sets

The shared-predictor comparison uses 2048-bit radius-two Morgan fingerprints. The historical source allocation is 60% model fitting, 20% stack fitting, and 20% conformal calibration. The three predictors are logistic regression, a logistic stack over logistic regression/random forest/extra trees/Bernoulli naive Bayes, and an equal-logit combination. All uncertainty rules within a predictor use the same probabilities and eligible rows. Development fits use seeds 101–105 and EPA fits 211–215.

For candidate label $y \in \{0,1\}$ and fitted probability $p_y(x)$, the least-ambiguous-classification score is $s(x,y) = 1 - p_y(x)$. For n_y observed class-y calibration scores, the ordinary Mondrian threshold is order statistic $k_y = \lceil (n_y+1)(1-\alpha_y) \rceil$, or $+\infty$ when $k_y > n_y$. An empty calibration class also receives an infinite threshold. The candidate label selects the threshold. The set contains y if $s(x,y) \le q_y$. The primary level is $\alpha_0 = \alpha_1 = 0.10$; levels 0.05 and 0.20 and deterministic adaptive-prediction-set scores are sensitivities [5].

We map $\{0\}$ to a negative/inactive decision, $\{1\}$ to a positive/active decision, and $\{0,1\}$ to deferral. Empty sets are repaired to the full set for all operational expansion comparisons; their raw frequency is retained separately. Let $S_0(x)$ denote this repaired baseline.

2.3 Transport expansion and its activation condition

TAME estimates target-to-calibration density ratios in two views: nine physicochemical descriptors and component/stack logits. Logistic domain-classifier odds are adjusted for domain sample sizes and clipped to $[0.05, 20]$. A third fingerprint view is retained as a diagnostic comparator. Sample sizes, effective sample sizes, clipping, domain discrimination, and weighted mean discrepancy determine view eligibility. The fixed gates and their one-at-a-time sensitivities are in Supplementary Table S2.

For a given view v, positive calibration weights w_{iv} and query weight $w_v(x)$ define the class distribution

$$F_{v,x,y} = \frac{\sum_{i:Y_i=y} w_{iv}\delta_{s(X_i,y)} + w_v(x)\delta_{+\infty}}{W_{v,y} + w_v(x)}, \qquad W_{v,y} = \sum_{i:Y_i=y} w_{iv}. \tag{1}$$

Its $(1-\alpha_y)$ quantile gives the weighted threshold. A protected label omitted by $S_0(x)$ is added if both required views pass the gates and contain that label. Otherwise the baseline is retained. We distinguish historical protection of label 0 from the prespecified label-1 objective of reducing erroneous inactive decisions for active compounds.

Define the strict lower-score mass

$$B_{v,y}(s) = \sum_{i:Y_i=y} w_{iv}\mathbf{1}\{s(X_i,y) < s\}. \tag{2}$$

The weighted set contains a finite candidate score $s = s(x,y)$ exactly when

$$B_{v,y}(s) < (1-\alpha_y)\{W_{v,y} + w_v(x)\}. \tag{3}$$

Equivalently, the query weight must exceed $B_{v,y}(s)/(1-\alpha_y) - W_{v,y}$. The strict inequality accounts for ties at a quantile boundary. With no class-y calibration observations it holds for every positive query weight, consistently with an infinite threshold. Supplementary Section S1 proves the condition and tests its boundary behavior against the executable quantile.

Equation 3 separates three events: view eligibility, addition of a label omitted by the baseline, and recovery of an erroneous decision. Each event is counted explicitly in the molecular evaluation.

If each view's query weight is constant across the target batch, its candidate threshold is constant. Under binary LAC, adding label 1 then selects an upper level set of $p_1(x)$ within baseline $\{0\}$ decisions. The two-view intersection uses the larger probability cutoff, and the union the smaller. At the resulting expansion count, ranking the omitted-label probability selects exactly the same set. Label-0 protection has the analogous result for $p_0(x)$ within baseline $\{1\}$. Query-varying weights are therefore necessary for a different ranking within this setting. Class-budget matching measures whether that difference improves error targeting.

2.4 Matching budgets and decomposing gains

For an expansion S of S_0, define

$$m_c = \sum_{i\in\mathscr{E}} \mathbf{1}\{S_0(x_i) = \{c\}, |S(x_i)| = 2\}, \qquad c \in \{0,1\}, \tag{4}$$

where $\mathscr{E}$ includes every eligible target, including those without an observed assay outcome. Each comparator expands exactly m_0 and m_1 decisions, ranked within their original singleton classes. The primary comparator ranks $p_{1-c}(x)$, the probability of the omitted label. Secondary rankings use uncertainty around 0.5, lower maximum training-set Tanimoto similarity, and fixed seeded random priorities. Stable molecule identifiers

resolve ranking ties. Target outcomes are absent from ranking and budget selection.

Within one baseline class-c pool, let R count expanded labeled decisions whose true class is $1-c$, and L count those whose true class is c. For the fixed labeled-class denominators N_y,

$$\Delta C_{1-c} = R/N_{1-c}, \qquad \Delta \mathrm{CSY}_c = -L/N_c, \qquad R+L = m_c^{\mathrm{obs}}. \tag{5}$$

Here C_y is class coverage and CSY_y is correct-singleton yield. The observed-label expansion count m_c^{obs} may differ from the operational count m_c when outcomes are missing. At equal observed-label budgets, greater error recovery necessarily preserves more correct singletons in the original class. At equal operational budgets, the observed-budget difference must also be included. Supplementary Section S1 gives this correction for two competing rules.

We report minimum-class coverage $C_{\min} = \min(C_0, C_1)$, macro correct-singleton yield $\mathrm{MacroCSY} = (\mathrm{CSY}_0 + \mathrm{CSY}_1)/2$, false-inactive exposure $E_1 = \Pr\{S = \{0\} \mid Y = 1\}$, false-active exposure $E_0 = \Pr\{S = \{1\} \mid Y = 0\}$, and decision yield. Operational deferral uses all eligible targets; coverage and errors use observed labels. Empty denominators remain undefined.

2.5 Comparisons and inference

The fixed primary contrast compares active-label TAME with its omitted-label-probability comparator for E_1, using the stack, LAC, $\alpha = 0.10$, and the historical six assays. The historical label-0 comparison explains the earlier coverage gain. The full matrix includes the three classical predictors, marginal/Mondrian LAC and APS, weighted variants, protected-label variants, single-view and union rules, ungated rules, fourteen gate sensitivities, and five gated plus five ungated query-weight permutations. Each expansion variant receives its own class-budget-matched probability comparator. Fixed workload curves use fractions 0, 0.10, 0.25, 0.50, 0.75, and 1 of the baseline singleton pool.

The end-to-end reference systems are independent paper-based CoDrug and KMM-CP implementations with AttentiveFP predictors. CoDrug includes its energy penalty, auxiliary molecular pool, and logit-space density estimates. Both printed and prespecified swapped energy-margin settings are retained. KMM-CP includes ordinary and selective kernel mean matching, bandwidth selection, and refitting after target selection. Candidate-label and literal predicted-group threshold interpretations are reported separately. Unselected targets defer in the full-cohort comparison. These systems use their own predictors and native workloads; their recorded settings and all ten configurations appear in Supplementary Section S3 and Table S4.

The experiment contains 960 completed predictor units: 360 development, 360 EPA, 120 random holdout, and 120 scaffold holdout. Of these, 480 use classical predictors and 480 use neural reference systems. Alpha levels and post-prediction variants reuse a unit's probabilities. Metrics are computed within assay–seed cells and then averaged equally over the declared panel. Bootstrap intervals resample molecular/structure identities jointly across assays and methods while retaining every seed prediction, using 2,000 replicates and seed 44021. The interval estimand is the fixed-panel mean; training seeds do not add independent molecules. Secondary intervals are pointwise exploratory.

Synthetic mechanisms with thirty replicates each examine no shift, overlapping covariate shift, support

loss, and changed $Y \mid X$ with identical observable covariates. These analyses are reported separately in Supplementary Section S5.

3 Results

3.1 Class-budget matching identifies a decision-aligned reference

On the historical six-assay EPA panel, omitted-label probability ranking achieved 85.10% minimum-class coverage and 46.57% MacroCSY, compared with 84.75% and 44.61% for generic probability uncertainty at the same class-specific expansion budget and 44.02% operational deferral. Similarity and random rankings also yielded lower coverage and MacroCSY (Table 1). Matching workload therefore exposes the practical value of ranking the label that can repair the original decision.

At this operating point, label-0 TAME increased mean minimum-class coverage from 83.739% to 85.104%, a gain of 1.3649 percentage points. Operational deferral increased by 2.0164 points, from 42.007% to 44.023%, and MacroCSY fell by 1.6067 points, from 48.179% to 46.572%. The expansion recovered false-active decisions while withholding additional correct positive singletons.

The omitted-label rule reached the same mean minimum-class coverage, MacroCSY, and deferral as TAME. The TAME-minus-probability difference in minimum coverage was 0.0000 points, with a molecule-cluster bootstrap 95% interval of $[-0.0279, 0.0287]$ points. Thus, the historical coverage gain is attainable using predictor probabilities alone at the same class-specific workload.

Table 1: Historical six-assay EPA means with the stack and $\alpha = 0.10$. Entries are percentages. The four simple ranking rules use the exact label-0 TAME class budgets. The last row uses the separate active-label objective. E_0 is false-active exposure.

Rule	$C_{\min}$	E_0	MacroCSY	Deferral
Mondrian LAC baseline	83.74	14.64	48.18	42.01
TAME, label 0	85.10	12.93	46.57	44.02
Omitted-label probability	85.10	12.93	46.57	44.02
Probability uncertainty	84.75	13.44	44.61	44.02
Chemical similarity	84.83	13.21	45.65	44.02
Seeded random selection	84.87	13.24	45.34	44.02
TAME, label 1	83.74	14.64	48.18	42.01

Figure 1 displays the assay-level results, including the six assays outside the historical panel.

3.2 Gate passage and active-label expansion separate

In the fixed active-label primary comparison, both transport views passed in every one of the 30 assay–seed cells. Nevertheless, the expansion count was zero in all 30. TAME, its matched comparators, and repaired Mondrian therefore produced the same primary decisions: false-inactive exposure was 10.532%, minimum-class coverage 83.739%, and operational deferral 42.007%. Zero intervention explains the zero matched contrast in this panel.

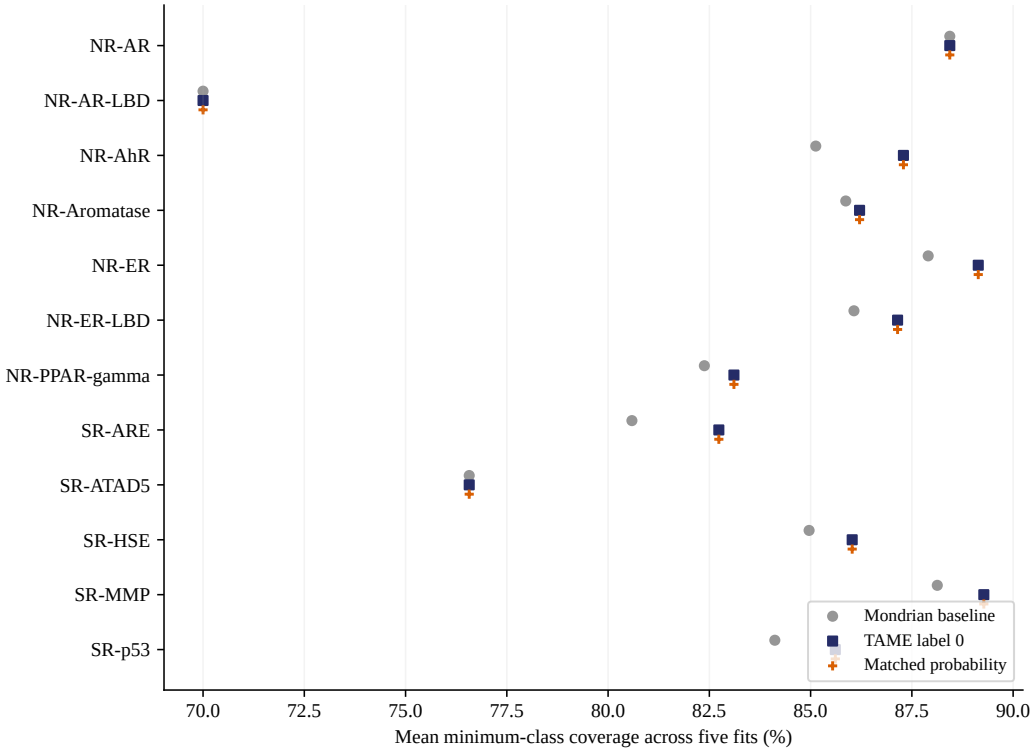


Figure 1: Assay-level EPA coverage for the stack at $\alpha = 0.10$, averaged over five fits. Markers are vertically offset for legibility.

Across all twelve EPA assays, both views passed in all 180 classical predictor cells. For the stacked predictor, active-label TAME added 21 compound–fit deferrals, concentrated in NR-AR and NR-AR-LBD. Seven were selected only by TAME and seven only by the matched probability rule. Neither rule recovered a false-inactive decision among these additions. Table 2 reports these compound–fit counts.

Table 2: Active-label mechanism counts at $\alpha = 0.10$ with the stack. Pass requires both views. Added counts are equal for TAME (A) and matched probability (P); A-only/P-only denotes discordant selections. Rescued lists erroneous inactive decisions recovered by A/P. Counts sum over fit seeds; the two EPA rows overlap.

Setting	Pass / cells	Added	A-only / P-only	Rescued
EPA, historical six	30 / 30	0	0 / 0	0 / 0
EPA, all twelve	60 / 60	21	7 / 7	0 / 0
AMES / random	5 / 5	0	0 / 0	0 / 0
CYP2D6 / random	5 / 5	0	0 / 0	0 / 0
DILI / random	0 / 5	0	0 / 0	0 / 0
hERG / random	0 / 5	0	0 / 0	0 / 0
AMES / scaffold	5 / 5	0	0 / 0	0 / 0
CYP2D6 / scaffold	5 / 5	19	0 / 0	1 / 1
DILI / scaffold	0 / 5	0	0 / 0	0 / 0
hERG / scaffold	0 / 5	0	0 / 0	0 / 0

3.3 Additional assay families distinguish fallback from nonactivation

AMES and CYP2D6 passed both transport gates in every shared-predictor fit under both split regimes. hERG failed the 100-calibration-observation threshold. DILI also had fewer than 100 target structures. Their active-label rules consequently retained the baseline.

For the stack, all four random-holdout tasks had zero additional deferrals. The scaffold CYP2D6 analysis added 19 compound–fit deferrals, identical under TAME and matched probability; both recovered one false-inactive decision. Its false-inactive exposure decreased from 10.410% to 10.369%, with deferral increasing from 34.375% to 34.520%. The remaining stacked scaffold tasks added no deferrals.

Across all 24 endpoint/split/predictor active-label comparisons, 23 had zero error differences. The remaining CYP2D6 random-split equal-logit contrast was -0.0418 points, with a pointwise interval of $[-0.1330, 0.0000]$. The experiment therefore distinguishes small-sample fallback, eligible but inactive expansion, and a small nonempty expansion reproduced by the simple ranking rule.

3.4 Predictor, workload, and reference-system comparisons

The underlying predictor changes the trade-off even before transport expansion. On the historical panel, ordinary equal-logit Mondrian had 84.82% minimum-class coverage and 46.70% MacroCSY at 44.37% deferral; logistic Mondrian had 84.84%, 43.51%, and 47.58%, respectively. The stack had lower minimum coverage but higher MacroCSY and fewer deferrals (Supplementary Table S3).

The fixed-workload curves show how coverage can rise as additional decisions are withheld (Figure 2). At its native operating point, deterministic Mondrian APS reached 91.38% minimum-class coverage and zero false-inactive exposure, while deferring 89.68% of eligible targets and retaining 9.28% MacroCSY. This operating point trades higher coverage for fewer informative decisions.

The neural reference systems likewise occupied different operating points. The swapped-margin CoDrug candidate-class variant gave 10.93% false-inactive exposure at 36.58% deferral and 52.97% MacroCSY. Selective KMM with candidate-label thresholds gave 7.67% false-inactive exposure at 52.79% deferral and 36.27% MacroCSY. All ten declared reference configurations, including the printed-margin CoDrug and predicted-group KMM interpretations, appear in Supplementary Table S4. These complete-system comparisons describe the trade-offs of different predictors and uncertainty rules.

3.5 Component ablations and controlled shift mechanisms

Removing gates did not activate the stack's primary active-label expansion. A physicochemical-only view or view union added sixteen decisions in one primary assay–seed cell; their own-budget error difference was -0.0952 points, with an interval of $[-0.3030, 0.0000]$. Across 280 declared own-budget ablation and query-alignment contrasts, no pointwise error interval excluded zero. The five gated and five ungated query-weight permutations each produced zero primary-panel budget, making that panel uninformative about the value of query-specific alignment.

In the synthetic paired no-shift and changed-conditional scenarios, all thirty pairs shared identical prediction and transport-diagnostic inputs. Active-label TAME nevertheless had mean class-1 coverage of

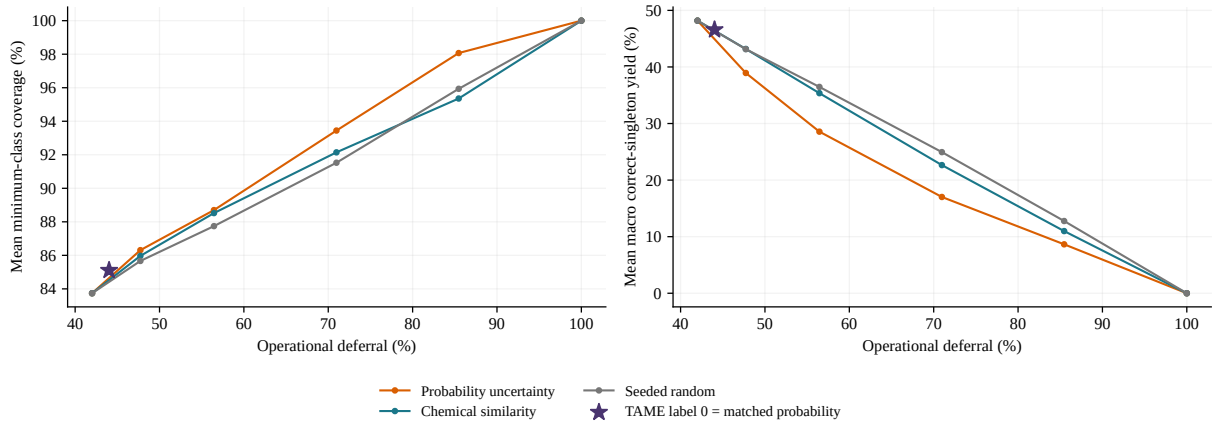


Figure 2: Coverage and correct-singleton yield across fixed additional-deferral fractions on the historical EPA panel. The star marks the label-0 TAME and matched-probability point. They share these aggregate coordinates. Curves use uncertainty, similarity, or random ranking over the baseline singleton pool; they describe workload trade-offs rather than class-budget-matched effects at every point.

90.36% and 39.74%, respectively, with the same 30.30% deferral. This mechanism check illustrates why diagnostics based only on the observed covariates cannot identify a changed conditional outcome law.

4 Discussion

Class-budget matching provides a way to measure the decision information in prediction-set expansion. Fixing expansion counts within the two original singleton classes separates the choice of compounds from the amount and class allocation of deferral. In the molecular case study, this comparison identifies an effective simple reference: omitted-label probability ranking achieves higher coverage and correct-singleton yield than generic uncertainty at the historical TAME budget. It also reproduces the 1.3649-point minimum-coverage gain using the predictor's probabilities alone.

The activation condition connects this comparison to the mechanism of weighted expansion. View eligibility, label addition, and error recovery are distinct events. The primary active-label rule passed both gates but added no labels; the all-twelve-assay rule made a small number of additions without recovering false-inactive errors. Counting these events localizes where a procedure changes molecular decisions and where it stops. It also gives a concrete basis for modifying an expansion rule: identify which candidate scores cross the weighted boundary, then measure the errors recovered by those additions.

The constant-query-weight result identifies the source of any departure from probability ranking under binary LAC. With fixed query mass, source weights and cell-level gates alter the threshold and budget while preserving the within-pool order. Query-dependent thresholds can change that order. Their contribution can therefore be tested directly by comparing recovered errors at the same class-specific workload. This connects transport estimation to the decision benefit it is intended to provide.

The empirical comparisons concern the declared assay panels and molecular cohorts. The shared-structure bootstrap accounts for compounds reused across fits and endpoints. Random and scaffold holdouts

extend the evaluation to four additional assay families, with sample-size fallback in hERG and DILI. The synthetic changed-conditional example identifies a separate limitation: identical observable transport inputs can accompany different outcome laws, so covariate diagnostics alone cannot determine coverage under that change.

For molecular screening, the resulting evaluation has three linked outputs: errors recovered, correct decisions withheld, and the workload of unresolved compounds. These quantities make the value of a coverage gain interpretable and provide a common basis for deciding when a transport-based expansion adds useful information.

5 Conclusion

Class-budget matching isolates error targeting from the volume and class allocation of deferral, and the exact activation condition links transport weights to label addition. In the molecular evaluation, omitted-label ranking improves the observed coverage–yield trade-off over generic uncertainty at the same workload and reproduces the historical TAME coverage gain. The inactive primary active-label result shows why eligibility and intervention must be reported separately. Together, these results provide a practical framework for evaluating the decision value of molecular prediction sets.

Data and code availability

The data adapters, experiment suite, frozen aggregate inputs, manuscript sources, and rebuilding scripts are available in the versioned project directory. The complete numerical supplement retains all returned cell metrics, panel metrics, and paired contrasts. The manuscript rebuilding script operates on frozen aggregates; retraining and molecule-level inference use the experiment suite and its full exports.

Declaration of competing interest

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CRediT authorship contribution statement

Siyuan Tong and Yuechen Wang contributed equally to this work and share first authorship.

Siyuan Tong: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. Yuechen Wang: Methodology, Validation, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies

OpenAI ChatGPT was used for language editing, manuscript organization, LaTeX formatting, code assistance, and preparation of reproducibility materials. The authors are responsible for the content of the submitted manuscript.

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